

EXPLORATORY DATA ANALYSIS AND VISUALISATION OF NATURAL LANGUAGE DATA

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1. INTRODUCTION

This report provides an analysis of text data from 26 fiction books by six English-language authors, downloaded from the Project Gutenberg website[1] as text files. I first perform an exploratory data analysis and visualisations of a subset of the data, chosen for diversity of writing style, before performing N-gram analysis across the corpus of all 26 documents (books). Finally, I use different dimension reduction and clustering methods for assessing the similarity of language used across the corpus.

2. DATA PREPROCESSING

Every text file in the data contains some redundant text specific to the Project Gutenberg website[1] at the start before asterisks, so I removed this portion of the text before analysing the text files. I additionally tokenized the data by removing punctuation and stop words as these words provide minimal insight into theme and writing style, and we are not interested in contractions since I do not perform sentiment analysis in this report.

3. EXPLORATORY ANALYSIS OF WRITING STYLE ACROSS AUTHORS

In this section, I explore a subset of the data containing one book from each of the three authors in the corpus with the most diverse writing styles. ChatGPT[2] gave me some nice phrases to describe the authors' general individual writing styles, along with the book in the dataset that is most typical of their writing style, shown in Table 1 below.

Author	Writing Style	Book Typical of Style
H.G. Wells	Imaginative and visionary	The Time Machine
Charles Dickens	Vivid and colourful descriptions	David Copperfield
Philip K. Dick	Mind-bending and thought-provoking	Piper in the Woods

Table 1: English-language authors and their writing styles, according to ChatGPT[2]

In the subsections below, I explore comparisons of the Term Frequency Inverse Document Frequency (TFIDF) of individual words, common bigrams, pairwise correlations between words and the relationship between term frequency and rank of word, which are common exploratory techniques in text analysis.

3.1 Term Frequency Inverse Document Frequency (TFIDF)

I calculated the words with the top 25 TFIDFs for each book in Table 1, using the definition

$$TFIDF = term\ frequency \cdot \log \left(\frac{number\ of\ documents}{document\ frequency + 1} \right)$$

However, the majority of these words were character names or place names. Whilst this reflects the nature of fiction books, it provides less information on the writing style of the author, so I repeated the analysis, excluding proper nouns and multiple non-informative words such as 'said' and 'could' from the text files. The results are shown in Figures 1 - 3.

Some similarities with the described writing styles can be seen in high-ranking TFIDF words. For example, 'sphinx' - a mythical creature - in 'The Time Machine' ties in with the imaginative style of Wells. Similarly, the set for 'Piper in the Woods' features diverse words such as 'asteroid', 'corporal' and 'electronic'. One wonders how these words all tie together which supports the thought-provoking description given.



Figure 1: Top 25 TF-IDF words in 'The Time Machine', excluding proper nouns and uninformative words, size-scaled by TFIDF



Figure 2: Top 25 TF-IDF words in 'Piper in the Woods', excluding proper nouns and uninformative words, size-scaled by TFIDF

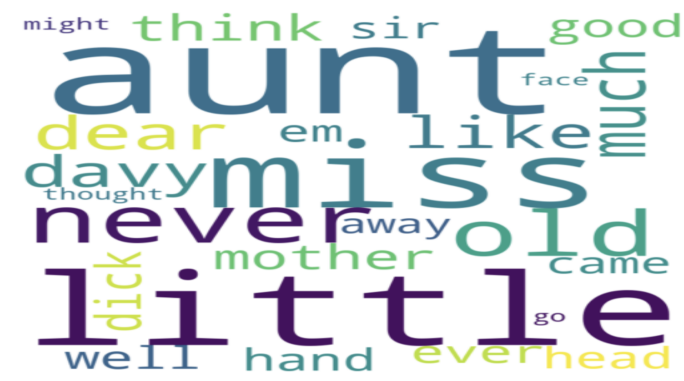


Figure 3: Top 25 TF-IDF words in 'David Copperfield', excluding proper nouns and uninformative words, size-scaled by TFIDF

The high TFIDF words for 'David Copperfield' did not tie in with the 'vivid and colourful' description but the aunt is clearly a very prominent figure (Figure 3). As found in the next section, some colourful names were omitted due to my design decision.

3.2 Bigrams

Figures 4 - 6 show common bigrams for the three books. For 'The Time Machine', the most interesting common bigram is 'little people', which conjures images of mythical minute humans or creatures, again emphasising the 'imaginative and visionary' style. The choice of the bigram 'medical man' rather than doctor also suggests a sense of magic around his healing which further plays to this theme. We can tell from the common bigrams for 'David Copperfield' that there are a lot of characters in the book from the number of names. Some of the names given are quite colourful, for example, 'Miss Trotwood'. There is clearly a military theme in 'Piper in the Woods', with bigrams like 'Commander Cox' and 'garrison chief'. Another notable bigram from this book is 'outer space', and this also ties in with the 'mind-bending and thought-provoking' description of Dick's writing-style, as one wonders how military and outer space tie together, as alluded to with 'asteroid' earlier.

3.3 Pairwise Correlation

'Sphinx', 'psychologist' and 'asteroid' were the most interesting words encountered, so a phi correlation analysis was conducted on these words to find the words most associated with them in the corpus. Limited information had been gained from 'David Copperfield', so I added 'aunt', one of the top TFIDF words, to the list. These are shown in Figure 7.

Interestingly, both negative and positive words are highly associated with 'aunt': 'admiring', 'error', and 'contradict'. According to ChatGPT[2], there are two primary aunts, Betsey Trotwood and Clara Peggotty, who are both admiring of David, whilst Clara Peggotty is also described as 'contradictory' and having 'errors in judgement'. The word associated with 'psychologist' that stands out is 'banged', and the two words together conjures an image of a character with anger or mental health issues. 'Asteroid' is associated with some very niche words ('eucalyptus' and 'tiberius'), suggesting that niche words tend to occur together. The generally high phi coefficients for these top 6 words indicates strong relationships between these words.

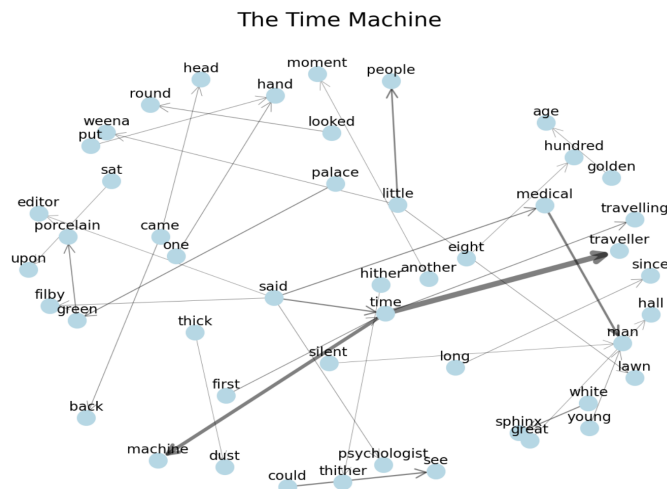


Figure 4: Bigrams occurring more than 4 times in 'The Time Machine'. Wider arrows indicator more occurrences.

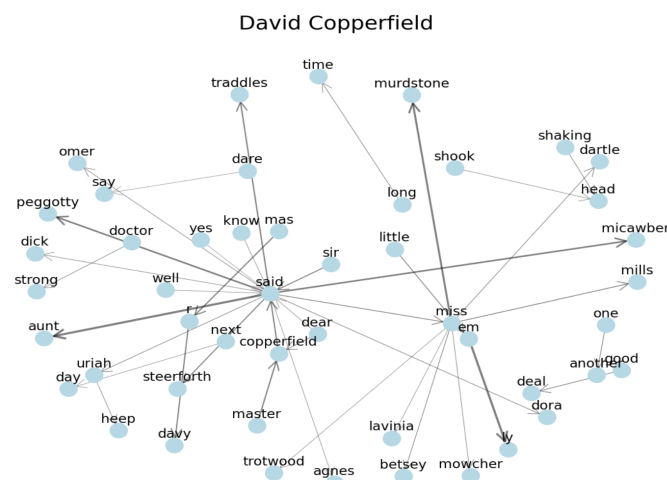


Figure 5: Bigrams occurring more than 40 times in 'David Copperfield'. Wider arrows indicator more occurrences.

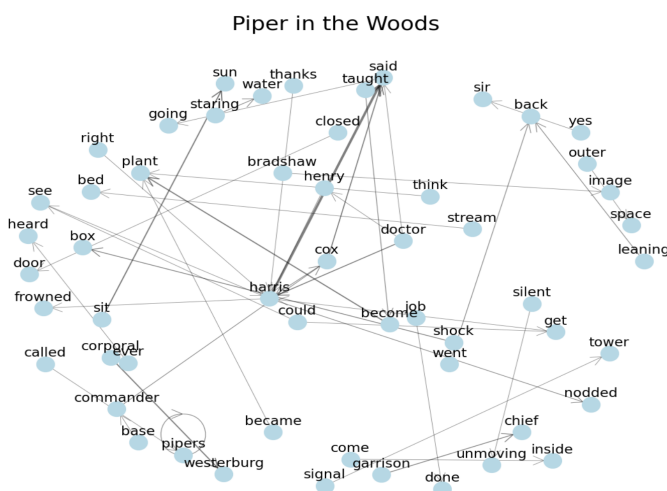


Figure 6: Bigrams occurring more than 2 times in 'Piper in the Woods'. Wider arrows indicator more occurrences.

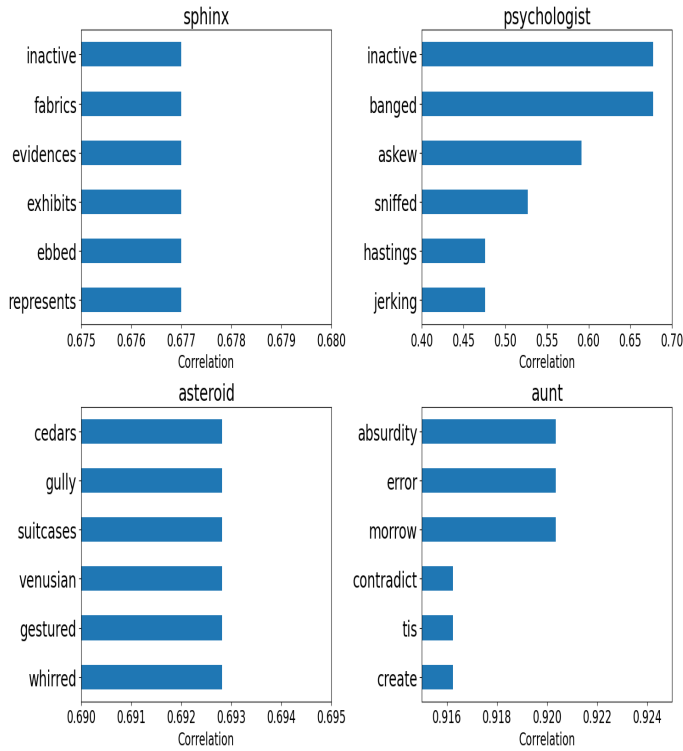


Figure 7: Phi Correlation coefficients for the top 6 words associated with 'sphinx', 'psychologist', 'asteroid' and 'aunt' in the corpus

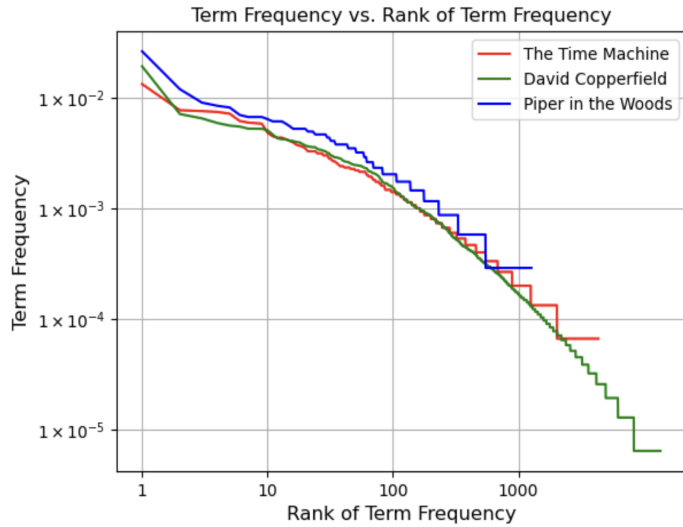


Figure 8: log-log co-ordinates plot of term frequency against rank of term frequency for each of the three books in Table 1

Book	The Time Machine	David Copperfield	Piper in the Woods
Slope	-0.86	-1.25	-0.77
95% C.I.	(-0.867, -0.856)	(-1.255, -1.247)	(-0.785, -0.763)

Table 2: Regression slopes for log(term frequency) against log(rank). Zipf's law suggests a slope of 1 generally holds.

3.4 Term Frequency versus Rank

Zipf's law states that the frequency that a word appears is inversely proportional to its rank, implying a slope of -1 in a regression of $\log(\text{term frequency})$ against $\log(\text{rank})$. From Table 2, we can see that whilst none of the 95% confidence intervals for these three books contained 1, the average of the three slope estimates is -0.96, which is very close, suggesting that the law holds on average. Figure 8 shows that Dick has the least variation in language in 'Piper in the Woods'; he uses the most common words more frequently and uses far fewer words than both Wells and Dickens (though the latter may reflect book length).

4. ASSESSING LANGUAGE SIMILARITY ACROSS BOOKS

In this section I explore the utility of different dimension reduction and clustering methods, for assessing the similarity of language across the corpus. I applied these methods to a (26 x 3022) feature matrix, containing the frequency of occurrence of the 200 most common bigrams for each of the 26 books across the corpus. I selected bigrams rather than higher N-grams due to better recall.

4.1 Dimension Reduction

Dimension reduction of a high-dimensional dataset makes it easier to visualise data and identify clusters, while reducing noise and improving computational efficiency. Below I compare two methods of dimension reduction: Principal Components Analysis and t-Distributed Stochastic Neighbor Embedding (t-SNE).

4.1.1 Principal Components Analysis (PCA). After performing PCA, I found an elbow at the first principal component, which explained around 30% of the total variance (see Figure 9). Subsequent components explained much smaller and gradually decreasing amounts of total variance. It took 15 components to capture 90% of the variability in the data, indicating that the structural relationships between variables are not easily captured in a small number of features. As PCA cannot capture enough variance with a small number of components, I explored t-SNE, which has the ability to capture non-linear relationships.

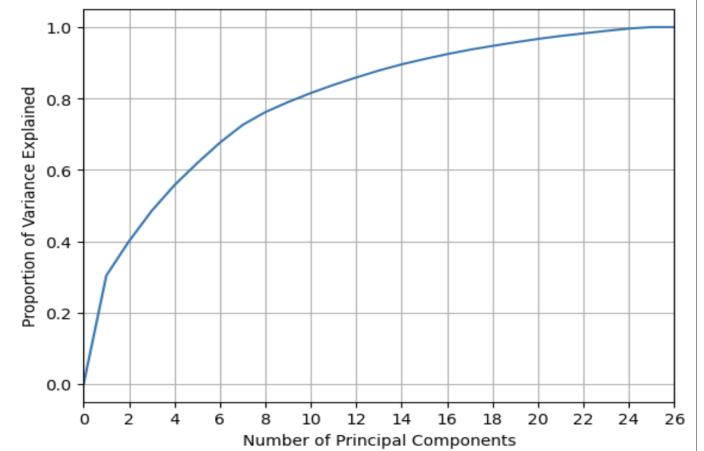


Figure 9: Proportion of variance explained for principal components of the high-dimensional matrix

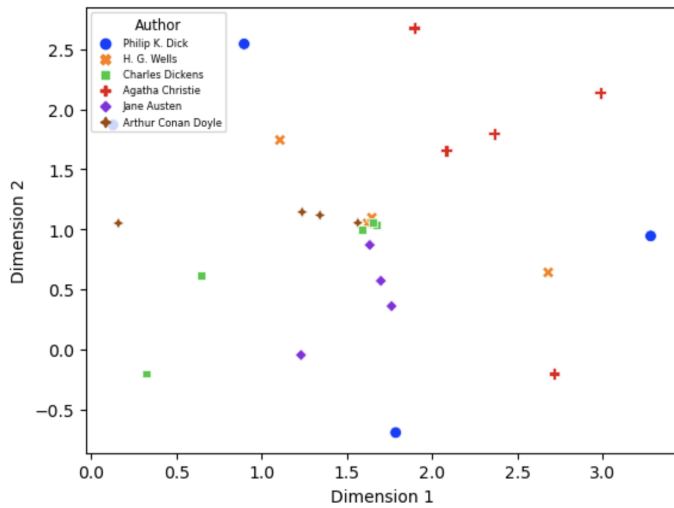


Figure 10: Projection of tSNE onto 2-dimensional embedding. Perplexity parameter of 12 and learning rate of 10.

4.1.2 t-SNE. I used PCA in the initialisation of t-SNE. Figure 10 shows the projection onto a two-dimensional embedding. I selected a perplexity parameter of 12 and learning rate of 10 after visualising cluster separability and picking the parameters with best separation. There is a clear central cluster, which indicates strong similarities in language - as reflected in a 2-dimensional representation of common bigram frequencies - between many of the works of Jane Austen, Charles Dickens, H.G. Wells and Arthur Conan Doyle. There also appears to be several weaker clusters (possibly around 5), with larger spacing between points than the central cluster. Notably, 4 of the 5 books by Agatha Christie lie close to each other, suggesting limited similarity in language with other books in the corpus by other authors.

The presence of clusters suggests that t-SNE has perhaps captured some non-linear relationships in the feature matrix that PCA has failed to capture. Since there was evidence of clustering in t-SNE, I opted to use the 2 dimensional t-SNE embedding in K-means clustering, which I detail below, in order to see if an algorithmic approach to clustering can mathematically define and specify groups.

4.2 K-Means Clustering

In order to find an optimal value of k , I calculated the Within-Cluster Sum of Squares (WCSS) for values of k from 1 to 10. Since the clusters (and the minimised value of the WCSS) can be sensitive to the initial (random) values that are allocated as cluster centres, I ran the k-means optimisation procedure 10 times (for each fixed k), each time using a different set of random starting points for the cluster centres. I additionally used a smart centroid initialization technique called K-Means++ as proposed by Arthur and Vassilvitskii (2007)[3], which chooses the first centroid uniformly at random from the data points and the subsequent centroids are chosen with a probability proportional to the squared distance from the nearest existing center. This spreads the cluster centroids more evenly across the data points, and can make clustering more accurate and efficient. As can be seen in Figure 11, there is an elbow in the WCSS graph at $k = 6$, so I used this optimal value of k in the algorithm.

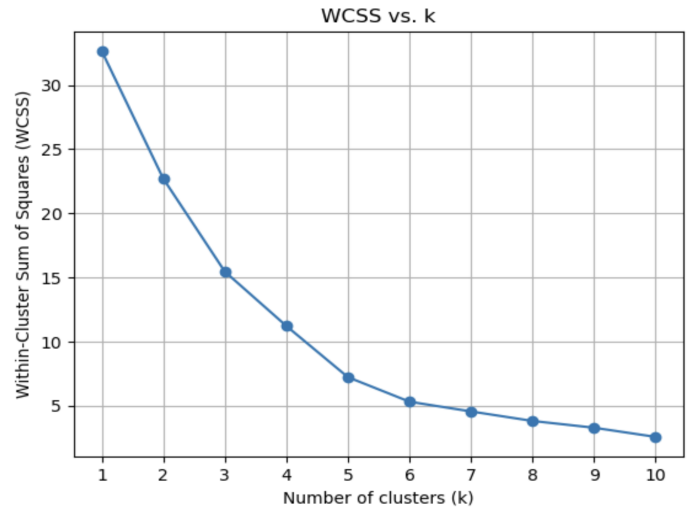


Figure 11: Within-Cluster Sum of Squares (WCSS) for various numbers of clusters applied to 2D t-SNE embedding

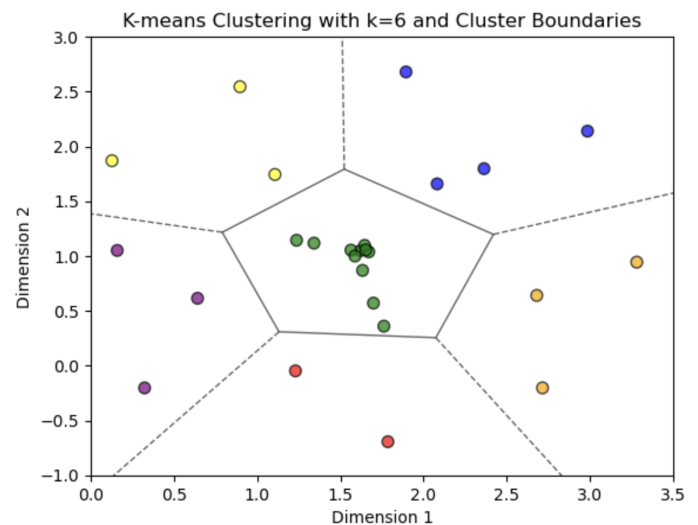


Figure 12: Voronoi diagram for k-means clustering ($k = 6$)

Figure 12 shows the application of k-means clustering, with $k = 6$, to the t-SNE projection of the bigram dataset onto a 2-dimensional embedding. We see that the groupings are those which you would naturally expect from looking at Figure 10. The Voronoi diagram defines the boundaries with black lines such that for any point in a bounded region, the distance to the centroid of that region's cluster is the minimum of all cluster centroids from that point. Whilst the central cluster is clearly the strongest, there are some looser clusters, implying loose relationships in language (specifically bigrams) between, for example, Jane Austen and Philip K. Dick (bottom cluster) or H.G. Wells and Philip K. Dick (top left cluster).

I investigated the central cluster further, as it contained a large proportion (42%) of the total books. Further analysis showed that this could be explained by historical context, book type and themes. The majority of books in the central cluster were written in the 19th century, a period which had a distinct literary style, which could have resulted in language similarities in terms of vocabulary, grammar and sentence structure. In contrast, Agatha

Christie and Dick were 20th century authors, a period which would have had a different linguistic writing style, and this is reflected in their books not lying within the central cluster. The books in the central cluster are also all considered to be classic literature, whereas Christie's writing style is generally less formal and more accessible to a broader audience, and Dick's style is 'imaginative and thought-provoking', as mentioned previously. According to ChatGPT[2], the themes in the central cluster also "tend to focus more on social issues, interpersonal relationships, and moral values", whereas those in the outer clusters "reach more diverse themes such as Christmas spirit in A Christmas Carol, satirical humor in The Pickwick Papers, and futuristic concepts in Philip K. Dick's stories". This also helps to explain the central cluster.

Only one of Christie's books, 'The Hunter's Lodge Case', lies in a cluster with other authors' books (bottom right cluster) so I was curious to find out what similarities in language it may have with these books. ChatGPT[2] indicated that the books in this cluster all share elements of suspense and mystery. There may perhaps be some similarities due to the type of language required to create tension. Christie's other detective fiction books may perhaps lie in a separate cluster due to these books being full-length novels, whereas 'The Hunter's Lodge Case' and 'The Crystal Crypt' (both in the bottom right cluster) are both short stories. Consequently, the more focused narratives and quicker resolutions may influence the language used.

CONCLUSION

I carried out an exploratory analysis of text data from 26 fiction books by six famous English-language authors. I first looked at a subset of three books with the most diverse writing styles, as described by ChatGPT[2], and analysed the TFIDF of individual words, common bigrams, pairwise correlations between words and the relationship between term frequency and rank of word. I found that the high-ranking TFIDF words generally tied in with the given ChatGPT descriptions of writing style, and bigrams further supported these descriptions, as well as giving insight into things such as theme and number of characters. Pairwise correlations was shown to be a useful tool to gain insight into characters such as the aunts in 'David Copperfield'. Zipf's law was found to hold on average across the subset of books.

I then explored dimension reduction and clustering methods for assessing the similarity of language across all 26 books. I found PCA did not capture the variability across the data with a small number of components, so I instead used t-SNE, and identified 6 clusters visually in a 2D projection embedding, which was confirmed as optimal by plotting WCSS in k-means clustering. After applying k-means clustering with $k = 6$, further research on a collection of 11 books in a strong central cluster showed commonalities in historical context, book type and themes. The other 5 clusters had much greater distance between the points but some plausible relationships in themes between the points in these clusters were identified and in particular, Agatha Christie shared the lowest similarity in language with other authors.

REFERENCES

- [1] *Project Gutenberg*. 2023. URL: <https://www.gutenberg.org/>.
- [2] *ChatGPT*. 2023. URL: <https://chat.openai.com/>.
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